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SARDAR PATEL UNIVERSITY
BCA Semester-4 Examination – April-2026 [OLD Course] {NC}
US04BCBA27 || OPERATING SYSTEMS

Date : 06-04--2026, Monday

Time : 02.00 p.m. – 05.00 p.m.



Total : 70 Marks
10

Q-1 Select the appropriate option for the following questions:

- 1) Process Monitoring in hospital is an example of which type of OS?
A. Multi User
B. Time Sharing
C. Real Time
D. None
- 2) A Program in execution is known as _____.
A. Process
B. Files
C. Commands
D. None of these
- 3) _____ is not a valid Memory allocation algorithm.
A. Suit-fit
B. Worse-fit
C. Next-fit
D. First-fit
- 4) A Lazy Swapper is also known as _____.
A. Pager
B. Frames
C. Pages
D. None of these
- 5) _____ is the solution of the bounded buffer.
A. Shared-Memory
B. Virtual memory
C. Data consistency
D. none
- 6) Which of the following is NOT the valid resource utilization state?
A. Request
B. Use
C. Execution
D. Release
- 7) _____ command is use to display help of LINUX commands.
A. helptop
B. mv
C. rm
D. man
- 8) Case statement ends with _____.
A. end
B. end case
C. esac
D. None of these
- 9) Time-sharing system also known as _____.
A. Real-time
B. Multi-tasking
C. Multi-user
D. None of these
- 10) Compiler is an example of which components of Operating system?
A. Hardware
B. System Program
C. Application Program
D. None

Q-2 Give short answers of the following questions: (Attempt TEN out of TWELVE) **20**

- 1) Define Long term scheduler.
- 2) Define Operating system.
- 3) Differentiate between process and program
- 4) Explain First-fit memory allocation techniques.
- 5) What is Belady's Anomaly?
- 6) What is Page fault?

(1)

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- 7) What is Mutual Exclusion?
- 8) When Race conditions arise?
- 9) Justify "Linux is a Secure Operating System".
- 10) Explain who command.
- 11) Explain cp command .
- 12) Explain man command .

- Q-3** A Explain Time Sharing Operating System. 06
B Which are the functions performed by Operating System? 04

OR

- Q-3** A Explain SJF scheduling algorithm in brief. 06
B Explain process state and PCB in brief. 04

- Q-4** A Explain FIFO Page replacement algorithm in detail. 05
B What is Fragmentation? List different types of Fragmentation. Explain in details. 05

OR

- Q-4** A Explain Swapping of two processes with suitable example. 05
B What is Demand Paging? Explain in brief. 05

- Q-5** A What is LINUX? Explain basic features of LINUX Operating System. 05
B What do you mean by Deadlock? Explain all necessary conditions for occurrence of deadlock. 05

OR

- Q-5** A What are the necessary conditions for Deadlock prevention? 05
B What is Cooperative Process? Explain Producer-Consumer Problem. 05

- Q-6** Explain grep command and chmod command in detail. [10]

OR

- Q-6** Explain all control structures in details. [10]

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